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Acknowledgement of Country

The Victorian Curriculum and Assessment Authority proudly acknowledges and pays respect to

Victoria's Aboriginal and Torres Strait Islander communities and their rich and enduring cultures.

We acknowledge Aboriginal and Torres Strait Islander people as Australia's first peoples and as the Traditional Owners and custodians of the lands and waters on which we rely. We pay respect to Elders past and present of the lands where we conduct our work and recognise their ongoing contributions as the first educators on the land now known as Victoria.

Mathematics

Curriculum

Level 10

Achievement standard

By the end of Level 10, students recognise the effect of approximations of real numbers in repeated calculations.

Students use mathematical modelling to solve problems involving growth and decay in financial and other applied situations, applying linear, quadratic and exponential functions as appropriate, and solve related equations, numerically and graphically. They make and test conjectures involving functions and relations using digital tools. Students substitute into formulas, find unknown values, manipulate linear and quadratic algebraic expressions, expand binomial expressions and factorise monic and simple non-monic quadratic expressions, with and without the use of digital tools. They solve problems involving linear equations and inequalities, quadratic equations and pairs of simultaneous linear equations and related graphs, algebraically and graphically, with and without the use of digital tools, and justify solutions. They represent linear, quadratic and exponential functions numerically, graphically and algebraically, and use them to model situations and solve practical problems. Students can design and implement simple algorithms using pseudocode or other general purpose programming language.

Students solve measurement problems involving surface area and volume of composite objects. They interpret and use logarithmic scales representing small or large quantities or change in applied contexts. Students apply Pythagoras' theorem and trigonometry to solve practical problems involving right-angled triangles. They identify the impact of measurement errors on the accuracy of results. Students use mathematical modelling to solve practical problems involving direct and inverse proportion and scaling, evaluating and modifying models, and reporting assumptions, methods and findings.

Students use deductive reasoning, theorems and algorithms to solve spatial problems. They interpret networks used to represent practical situations and describe connectedness.

Students compare univariate data sets by referring to summary statistics and the shape of their displays. They plan and conduct statistical investigations involving bivariate data, including where the independent variable is time. They represent the distribution of data involving 2 variables, using tables and scatterplots, and comment on possible association. They analyse inferences and conclusions in the media, noting potential sources of bias. Students compare the distribution of continuous numerical data, using various displays, and discuss distributions in terms of centre, spread, shape and outliers.

Students apply conditional probability to solve problems involving compound events. They design and conduct simulations involving conditional probability, using digital tools.

Content descriptions and elaborations

Strand: Number

Content descriptions	Elaborations
<i>Students learn to:</i> recognise the effect of using approximations of real numbers in repeated calculations and compare the results when using exact representations VC2M10N01	<i>This may involve students:</i> - comparing and contrasting the effect of truncation or rounding on the final result of calculations when using approximations of real numbers rather than exact representations - investigating the impact of approximation on multiple calculations in contexts that involve the area of compound shapes involving circles, the surface area and volume of compound objects, and repeated calculations of simple interest where the solutions are not exact cents

Strand: Algebra

Content descriptions	Elaborations
<i>Students learn to:</i>	<i>This may involve students:</i>

Content descriptions	Elaborations
factorise algebraic expressions by taking out a common algebraic factor VC2M10A01	• using the distributive law and the exponent laws to factorise algebraic expressions • understanding the relationship between factorisation and expansion
<i>Students learn to:</i> simplify algebraic products and quotients using exponent laws VC2M10A02	<i>This may involve students:</i> • applying knowledge of exponent laws to algebraic terms, and simplifying algebraic expressions using both positive and negative integral indices
<i>Students learn to:</i> apply the 4 operations to simple algebraic fractions with numerical or single variable denominators VC2M10A03	<i>This may involve students:</i> • expressing the sum and difference of algebraic fractions with a common denominator • using the exponent laws to simplify products and quotients of algebraic fractions
<i>Students learn to:</i> expand binomial products and factorise monic quadratic expressions using a variety of strategies VC2M10A04	<i>This may involve students:</i> • exploring the method of completing the square to factorise quadratic expressions and solve quadratic equations • identifying and using common factors, including binomial expressions, to factorise algebraic expressions using the technique of grouping in pairs • using the identities for perfect squares and the difference of squares to factorise quadratic expressions
<i>Students learn to:</i> substitute values into formulas to determine an unknown and rearrange formulas to solve for a particular term VC2M10A05	<i>This may involve students:</i> • solving simple equations arising from formulas • rearranging expressions to make a specified variable the subject, such as calculating the radius of a sphere to produce a given volume
<i>Students learn to:</i> implement algorithms that use	<i>This may involve students:</i> • using two-dimensional arrays such as matrices to represent and implement sequences of transformations

Content descriptions	Elaborations
data structures using pseudocode or a general purpose programming language VC2M10A06	of sets of points in the plane • using pointers in algorithms
<i>Students learn to:</i> solve problems involving linear equations, including those derived from formulas VC2M10A07	<i>This may involve students:</i> • representing word problems with simple linear equations and solving them to answer questions
<i>Students learn to:</i> solve linear inequalities and graph their solutions on a number line VC2M10A08	<i>This may involve students:</i> • representing word problems with simple linear inequalities and solving them to answer questions • graphing regions corresponding to inequalities in the Cartesian plane (for example, graphing $2x + 3y < 24$) and verifying using a test point such as (0, 0)
<i>Students learn to:</i> solve simultaneous linear equations, using algebraic and graphical techniques including using digital tools VC2M10A09	<i>This may involve students:</i> • investigating situations involving simultaneous linear equations in context (such as multiple quotes for a job, or profit and loss) and solving the equations graphically, giving solutions in everyday language (such as 'break-even point' or 'point to change providers' for the job) • describing the solution of simultaneous equations within the context of the situation • using the algebraic methods of substitution and elimination, with and without the use of digital tools, and identifying which technique is the most efficient for a given pair of equations • associating the solution of simultaneous equations with the coordinates of the intersection of their corresponding graphs • investigating the strategies inherent in Aboriginal and/or Torres Strait Islander children's instructive games, for example, Weme from the Warlpiri Peoples of central Australia, and their connection to strategies to solve simultaneous linear equations in 2 variables

Content descriptions	Elaborations
<i>Students learn to:</i> solve problems involving gradients of parallel and perpendicular lines VC2M10A10	<i>This may involve students:</i> • solving problems using the fact that parallel lines have the same gradient and conversely that if 2 lines have the same gradient then they are parallel • solving problems using the fact that the product of the gradients of perpendicular lines is -1 and conversely that if the product of the gradients of 2 lines is -1 then they are perpendicular
<i>Students learn to:</i> explore the connection between algebraic and graphical representations of relations such as simple quadratic, reciprocal, circle and exponential, using digital tools as appropriate VC2M10A11	<i>This may involve students:</i> • sketching graphs of parabolas, and circles • applying translations, reflections and stretches to parabolas and circles • sketching the graphs of exponential functions using transformations • plotting graphs of families of relations where the product of 2 variables is equal to a fixed constant
<i>Students learn to:</i> solve linear equations involving simple algebraic fractions VC2M10A12	<i>This may involve students:</i> • solving a wide range of linear equations, including those involving one or 2 simple algebraic fractions, and checking solutions by substitution • representing word problems, including those involving fractions, as equations and solving them to answer the question
<i>Students learn to:</i> solve simple quadratic equations using a range of strategies, including null factor law VC2M10A13	<i>This may involve students:</i> • using techniques to solve quadratic equations, including grouping like terms, completing the square, the quadratic formula and null factor law, and choosing 2 integers with the required product and sum • using digital tools to explore the conditions for solutions to exist • using digital tools to explore both graphically and algebraically the link between the value of the discriminant and the number of real solutions to a quadratic equation
<i>Students learn to:</i> solve simple exponential equations	<i>This may involve students:</i>

Content descriptions	Elaborations
VC2M10A14	• investigating exponential equations derived from authentic mathematical models based on population growth • recognising that in a table of values, if the ratio between consecutive values of the dependent variable is constant, then it is an exponential relation • investigating the links between algebraic and graphical representations of exponential functions using graphing software • using digital tools with symbolic manipulation functionality to systematically explore exponential relations • investigating Aboriginal and/or Torres Strait Islander ranger groups' and other groups' programs that attempt to eradicate feral animals for survival of native animals on Country/Place, exploring the competition between feral and native animals and their impact on natural resources by formulating exponential equations for population growth for each animal species
<i>Students learn to:</i> use mathematical modelling to solve applied problems involving inverse proportion, growth and decay, including in financial contexts to establish the compound interest formula as repeated applications of simple interest; formulate problems, choosing to apply linear, quadratic or exponential models; interpret solutions in terms of the situation; evaluate and modify models as necessary and report assumptions, methods and findings VC2M10A15	<i>This may involve students:</i> • modelling problems involving inverse proportion in real-life contexts such as travel time decreasing with increased travel speed, the brightness of luminous objects decreasing with distance, or household expenditure and savings, and solving related equations • modelling situations and choosing between linear, quadratic and exponential models by representing relationships in a table of values and recognising that linear functions have constant first differences, quadratic functions have constant second differences and exponential functions have a constant ratio between consecutive values of the dependent variable • modelling situations involving exponential growth and decay, and contrasting this with linear growth or decay, for example, situations involving constant percentage change and constant ratio, and determining doubling time and half-life and approximate intervals for which the values of the model lie within a given range • modelling situations that involve working with authentic information, data and interest rates to calculate

Content descriptions	Elaborations
	compound interest and solve related problems • modelling and investigating how exponential equations are used in carbon dating to estimate the age of Aboriginal and Torres Strait Islander artefacts or material culture • modelling situations involving the growth of native animal populations on Country/Place with varying reproductive behaviour, using exponential functions and critiquing their applicability to real-world situations
<i>Students learn to:</i> solve equations graphically or using systematic numerical guess-check-and-refine with digital tools, with consideration of whether all solutions have been found VC2M10A16	<i>This may involve students:</i> • refining intervals on graphs and/or in tables of values to determine with increasing accuracy when the values of 2 functions are approximately equal • applying the graphing zoom functionality of digital tools and systematically refining intervals to identify approximate location of points of intersection of the graphs of 2 functions, such as $x^2 = 2^x$

Strand: Measurement

Content descriptions	Elaborations
<i>Students learn to:</i> solve problems involving the surface area and volume of composite objects using appropriate units VC2M10M01	<i>This may involve students:</i> • determining the volumes and surface areas of composite solids, formed from a range of right prisms and cylinders, by considering the individual solids from which they are constructed • estimating the surface area and volume of composite objects in practical contexts • using mathematical modelling to provide solutions to problems involving surface area and volume; for example, ascertaining the rainfall that can be saved from a rooftop and the optimal shape and dimensions for rainwater storage based on where it will be located on a property, or determining whether to hire extra freezer space for the amount of ice cream required at a fundraising event for the school or community
<i>Students learn to:</i>	<i>This may involve students:</i>

Content descriptions	Elaborations
interpret and use logarithmic scales in applied contexts involving small and large quantities and change VC2M10M02	• understanding that the logarithmic scale is calibrated in terms of order of magnitude, for example, doubling or powers of 10 • identifying and interpreting data representations (charts and graphs) that use logarithmic scales (for example, graphs representing percentage change, a wide range of values or exponential growth) and discussing when it is appropriate to use this type of scale and when it is not appropriate • investigating and interpreting logarithmic scales used in real-world contexts (for example, Richter, decibel and sensitivity scales or growth in investments, timescales and the spread of microorganisms and disease) and describing reasons for choosing to use a logarithmic scale rather than a linear scale • investigating dating methods of geological sites to provide evidence of Aboriginal and Torres Strait Islander Peoples' presence in Australia, including the Madjedbebe dig in the Northern Territory, that use logarithmic scales, scientific notation and measurement accuracy in the dating
<i>Students learn to:</i> solve practical problems by applying Pythagoras' theorem and trigonometry to right-angled triangles, including problems involving direction and angles of elevation and depression VC2M10M03	<i>This may involve students:</i> • applying right-angled trigonometry to solve navigation problems involving bearings; for example, determining the bearing and estimating the distance of the final leg of an orienteering course • applying Pythagoras' theorem and trigonometry to problems in surveying and design, where three-dimensional problems are decomposed into two-dimensional problems; for example, investigating the dimensions of the smallest box needed to package an object of a particular length • using a clinometer to measure angles of inclination, and applying trigonometry and proportional reasoning to determine the height of buildings in practical contexts • applying Pythagoras' theorem and trigonometry and using dynamic geometry software to design three-dimensional models of practical situations involving angles of elevation and depression; for example, modelling a crime scene • exploring navigation, design of technologies or surveying by Aboriginal and Torres Strait Islander Peoples, investigating geometric and spatial reasoning and how these connect to trigonometry

Content descriptions	Elaborations
<i>Students learn to:</i> use mathematical modelling to solve practical problems involving direct and inverse proportion and scaling of objects; formulate problems and interpret solutions in terms of the situation, including the impact of measurement errors on the accuracy of results; evaluate and modify models as necessary, and report assumptions, methods and findings VC2M10M04	<i>This may involve students:</i> • exploring contexts where measurement errors may impact research results and how measurement data impacted by error can result in biased findings • using plans and elevation drawings to investigate making changes to building designs, employing appropriate scales and converting to actual measurements within the context to make decisions about changes • analysing and applying scale and ratios in situations such as production prototypes and 3D printing; for example, using a 3D printer to produce scaled versions of actual objects • estimating the scale of an object by measuring a linear dimension and using typical life-size dimensions to determine the scale factor; for example, measuring the dimensions of a toy car to determine a scale that can be used to estimate the dimensions of a life-size car • investigating compliance with building codes and standards in design and construction, such as for escalators in shopping centres

Strand: Space

Content descriptions	Elaborations
<i>Students learn to:</i> apply deductive reasoning to formulate proofs involving shapes in the plane and use theorems to solve spatial problems VC2M10SP01	<i>This may involve students:</i> • distinguishing between a practical demonstration and a proof; for example, demonstrating that triangles are congruent by placing them on top of each other, as compared to using congruence tests to establish that triangles are congruent • developing proofs involving congruent triangles and angle properties, communicating the proof using a sequence of logically connected statements • applying an understanding of relationships to deduce properties of geometric figures; for example, the base angles of an isosceles triangle are equal

Content descriptions	Elaborations
	- investigating proofs of geometric theorems and using them to solve spatial problems, for example, applying logical reasoning and similarity to proofs and numerical exercises involving plane shapes, and using visual proofs to justify solutions - using dynamic geometry software to find the quadrilateral that has a vertex on each side of a rectangle and has the shortest possible path, and proving that the path forms a parallelogram
<i>Students learn to:</i> interpret networks and network diagrams used to represent relationships in practical situations and describe connectedness VC2M10SP02	<i>This may involve students:</i> - investigating how networks and network diagrams can be used to model authentic situations, recognising what real-world quantity is represented by the nodes (vertices), and what real-world quantity is represented by the links between them (edges) - investigating the use of graphs to represent a network, for example, investigating the Seven Bridges of Königsberg problem - investigating how polyhedrons may be represented as a network using edges, vertices, and interior and exterior faces; and representing the number of edges, vertices and faces in a table, exploring and demonstrating how Euler's formula $F + V = E + 2$ applies - investigating how a social network, intranet, local area network (LAN), electrical wiring or wireless network of a home can be represented as a network diagram to specify relationships; for example, using network diagrams to investigate practical problems involving connections, power overload or the need for routers - investigating the use of networks to represent authentic situations, for example, rail or air travel between or within London, Paris and Hong Kong; a food web representing a simple ecosystem; metabolic networks and other chemical or biological structures - representing Aboriginal and Torres Strait Islander Peoples' kinship systems using network diagrams and exploring the significance of relationships to Country/Place

Content descriptions	Elaborations
<i>Students learn to:</i> compare data distributions for continuous numerical variables using quartiles and interquartile range and appropriate data displays including boxplots, histograms and dot plots; discuss the shapes of these distributions in terms of centre, spread, shape and outliers in the context of the data VC2M10ST01	<i>This may involve students:</i> • finding the five-number summary (minimum and maximum values, median and upper and lower quartiles) and using its graphical representation, the boxplot, as tools for both numerically and visually comparing the centre and spread of data sets • exploring the effect of varying data values, including outliers, on the interquartile range for different sets of data • constructing and interpreting boxplots and using them to compare data sets, understanding that boxplots are an efficient and common way of representing and summarising data and can facilitate comparisons between data sets • comparing shapes of distributions using boxplots, histograms, cumulative frequency graphs and dot plots, discussing symmetry, skew and modality • using digital tools to compare boxplots and histograms as displays of the same data in the light of the statistical questions being addressed and the effectiveness of the display in helping to answer the question • comparing the information that can be extracted and the stories that can be told about continuous and discrete numerical data sets that have been displayed in different ways, including histograms, dot plots, boxplots and cumulative frequency graphs
<i>Students learn to:</i> construct scatterplots and consider a line of good fit; comment on the association between the 2 numerical variables in terms of strength, direction and linearity VC2M10ST02	<i>This may involve students:</i> • discussing the difference between association and cause and effect, and relating this to situations such as health, diversity of species and climate • using statistical evidence to make, justify and critique claims about association between variables, such as in contexts of climate change, migration, online shopping and social media • informally using a line of good fit by eye and using digital tools to compare and discuss the reliability of any predictions • investigating the relationship between 2 variables of spearthrowers used by Aboriginal and Torres Strait

Content descriptions	Elaborations
	Islander Peoples by using data to construct scatterplots, make comparisons and draw conclusions
<i>Students learn to:</i> construct two-way tables and discuss possible relationship between categorical variables VC2M10ST03	<i>This may involve students:</i> • using two-way tables to investigate and comparing the survey responses to questions involving five-point Likert scale against 2 different categories of respondents, for example, junior students' responses compared to senior students' responses to a survey question • recording data in two-way tables and using percentages and proportions to identify patterns and associations in the data • conducting a litter survey around the school, considering the relationship between different categorical variables such as the day of the week (as canteen specials might lead to different types of litter) or the weather (due to hot days leading to more icy poles and cold drinks being sold)
<i>Students learn to:</i> analyse claims, inferences and conclusions of statistical reports in the media and other places, by linking claims to displays, statistics and representative data, including ethical considerations and identification of potential sources of bias VC2M10ST04	<i>This may involve students:</i> • identifying potentially misleading data representations in the media such as graphs with broken axes and scales that do not start at zero or are nonlinear; and recognising when data is not relating to the claim, not representative of the population or is deliberately being used to mislead or support a claim or biased point of view • investigating the source and size of the sample from which the data was collected and deciding whether the sample is appropriately representative of the population • investigating population proportions and rates, and discussing potential ethical considerations, for example, when presenting statistical data involving infection rates and the number of cases per head of population • discussing the ethical considerations associated with collecting data to predict the number of people likely to be infected with a strain of flu or experience side effects with a certain medication, considering validity claims and sample sizes • using the concept of Indigenous data sovereignty to critique and evaluate the Australian Government's

Content descriptions	Elaborations
	'Closing the Gap' report
<i>Students learn to:</i> plan and conduct statistical investigations of situations that involve bivariate data, including where the independent variable is time; evaluate and report findings with consideration of limitations of any inferences VC2M10ST05	<i>This may involve students:</i> - designing statistical investigations that collect bivariate data over time through observation, experiment or measurement; graphing, interpreting and analysing data; and reporting within the context of the statistical investigation question - investigating and comparing anecdotal claims with data collected from well-constructed surveys, including those concerning climate, housing affordability and natural resources, with consideration of data validity and limitations of interpolation or extrapolation - using a statistical investigation to address the question 'Is there a relationship between vaccines and immunity from a virus?' - investigating biodiversity changes in Australia before and after colonisation by comparing related bivariate numerical data, discussing and reporting on associations

Strand: Probability

Content descriptions	Elaborations
<i>Students learn to:</i> use the language of 'if ... then ...', 'given', 'of' and 'knowing that' to investigate conditional statements and identify common mistakes in interpreting such language, and describe and interpret situations involving conditional probability; design and conduct simulations using digital tools to model conditional probability and	<i>This may involve students:</i> - using two-way tables and Venn diagrams to understand conditional statements using the language of 'if ... then ...', 'given', 'of' and 'knowing that' and identifying common mistakes in interpreting such language - using arrays and tree diagrams to represent, interpret and compare probabilities of dependent and independent events - using samples of different sizes with and without replacement from a population to identify when the difference in methods becomes negligible - using simulations to gather data on frequencies for situations involving chance that appear to be counterintuitive, such as the three-door problem or the birthday problem

Content descriptions	Elaborations
interpret results VC2M10P01	- identifying situations in real life where probability simulations are used for decision-making, such as supply and demand of product, insurance risk and queueing - using simulation to predict the number of people likely to be infected with a strain of flu or virus
<i>Students learn to:</i> describe the results of two- and three-step chance experiments, both with and without replacements, assign probabilities to outcomes and determine probabilities of events; investigate the concept of independence VC2M10P02	<i>This may involve students:</i> recognising that a subsequent event can be dependent, and that this will affect the way its probability is calculated, for example, cards chosen from a standard deck without replacement, conditional selection problems from a fixed sample size, or the Bertrand's box paradox

